

Supply Chain Monitoring and Optimization Platform

Week 1 — Database Setup & Shipments Collection

Field	Details
Name	Vaishali.S
Week	Week 1 — Database Foundation
Technology	MongoDB / mongosh
Database	supply_chain
Collection	shipments
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Week 1 Objectives

- Set up a MongoDB database named supply_chain
- Create and populate the shipments collection with sample documents
- Insert multiple shipment records using insertMany()
- Create an index on order_id for optimized querying
- Verify data insertion using the find() method

Key Concepts Applied

6.1 insertMany()

The insertMany() method inserts multiple documents into a collection in a single operation. It is more efficient than calling insertOne() repeatedly and returns an acknowledgement object containing the auto-generated _id values.

6.2 createIndex()

Indexes in MongoDB improve the speed of query operations by reducing the number of documents MongoDB must scan. An ascending index (value: 1) on order_id allows efficient lookups and range queries on that field.

6.3 ObjectId

MongoDB automatically assigns a unique ObjectId to the _id field of each document upon insertion. ObjectIds are 12-byte identifiers that encode a timestamp, machine identifier, process ID, and a random counter.

Output

```
>_MONGOSH
> db.shipments.insertMany([
  {
    shipment_id:1,
    order_id:101,
    location:"chennai",
    status:"in transit"
  },
  {
    shipment_id:2,
    order_id:102,
    location:"bangalore",
    status:"delivered"
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a0547bdccdd47dc23108a52'),
    '1': ObjectId('6a0547bdccdd47dc23108a53')
  }
}
> db.shipments.createIndex({order_id:1})
< order_id_1
> db.shipments.find()
< {
  _id: ObjectId('6a0547bdccdd47dc23108a52'),
  shipment_id: 1,
  order_id: 101,
  location: 'chennai',
  status: 'in transit'
}
```

```
    status: 'in transit'
  }
  {
    _id: ObjectId('6a0547bdccdd47dc23108a53'),
    shipment_id: 2,
    order_id: 102,
    location: 'bangalore',
    status: 'delivered'
  }
shipment>
```